

Chapter 2, Problem 2P

Problem

Find the hot resistance of a light bulb rated 60 W, 120 V.

Step-by-step solution

Step 1 of 1

Consider the following data:

The value of the voltage, $V = 120 \text{ V}$

The power rating of the bulb, $P = 60 \text{ W}$

Write the expression for the power in terms of voltage and resistance.

$$P = \frac{V^2}{R}$$

Calculate the value of the hot resistance of the bulb.

$$\begin{aligned} R &= \frac{V^2}{P} \\ &= \frac{(120)^2}{60} \\ &= \frac{14400}{60} \\ &= 240 \, \Omega \end{aligned}$$

Therefore, the value of the hot resistance of the bulb, R is $\boxed{240 \, \Omega}$.